NAME: _____ Geometry (FL B.E.S.T.)

6.3 Sine, Cosine, and Tangent (SOH-CAH-TOA)

For an acute angle θ in a right triangle, label the sides relative to θ: the **hypotenuse** (always opposite the right angle), the **opposite** side (across from θ), and the **adjacent** side (next to θ, not the hypotenuse). The three basic trig ratios are $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$ — remembered by **SOH-CAH-TOA**. Because all right triangles with the same acute angle are similar, these ratios depend only on the angle, never on the triangle's size. A useful consequence: the two acute angles A and B of a right triangle are complementary ($A + B = 90°$), and the side opposite A is the same side that's adjacent to B — so $\sin A = \cos B$ always. This worksheet practises writing ratios, solving for missing sides, and combining trig with the Pythagorean Theorem.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Opposite $= 9$, hypotenuse $= 15$. Find $\sin\theta$.

2. Adjacent $= 8$, hypotenuse $= 17$. Find $\cos\theta$.

3. Opposite $= 5$, adjacent $= 12$. Find $\tan\theta$.

4. $\sin\theta = \dfrac{4}{5}$, hypotenuse $= 25$. Find the opposite side.

5. $\cos\theta = \dfrac{3}{5}$, hypotenuse $= 20$. Find the adjacent side.

6. $\tan\theta = \dfrac{5}{12}$, adjacent $= 24$. Find the opposite side.

7. $\sin\theta = \dfrac{8}{17}$, opposite $= 16$. Find the hypotenuse.

8. Opposite $= 6$, adjacent $= 8$, hypotenuse $= 10$. Find $\sin\theta$, $\cos\theta$, $\tan\theta$.

9. $\tan\theta = \dfrac{7}{24}$, opposite $= 14$. Find the adjacent side.

10. Adjacent $= 20$, hypotenuse $= 25$. Find $\cos\theta$.

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11. $\cos\theta = \frac{12}{13}$, adjacent $= 36$. Find the hypotenuse.

12. If $\sin A = \frac{5}{13}$, what is $\cos B$ for the other acute angle B?

13. A right triangle has legs 20 and 21. Find the hypotenuse, then $\sin\theta$ and $\cos\theta$ for the angle opposite the leg of 20.

14. $\tan\theta = \frac{3}{4}$. Find $\sin\theta$ and $\cos\theta$.

15. A right triangle has hypotenuse 50 and $\sin\theta = 0.6$ for one acute angle. Find the opposite side, then the adjacent side (Pythagorean Theorem), then $\cos\theta$.

✓ Answer Key

1. $\frac{3}{5}$
2. $\frac{8}{17}$
3. $\frac{5}{12}$
4. 20
5. 12
6. 10
7. 34
8. $\frac{3}{5}$; $\frac{4}{5}$; $\frac{3}{4}$
9. 48
10. $\frac{4}{5}$
11. 39
12. $\frac{5}{13}$
13. Hypotenuse $= 29$; $\sin\theta = \frac{20}{29}$, $\cos\theta = \frac{21}{29}$
14. $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$
15. Opposite $= 30$; adjacent $= 40$; $\cos\theta = \frac{4}{5}$